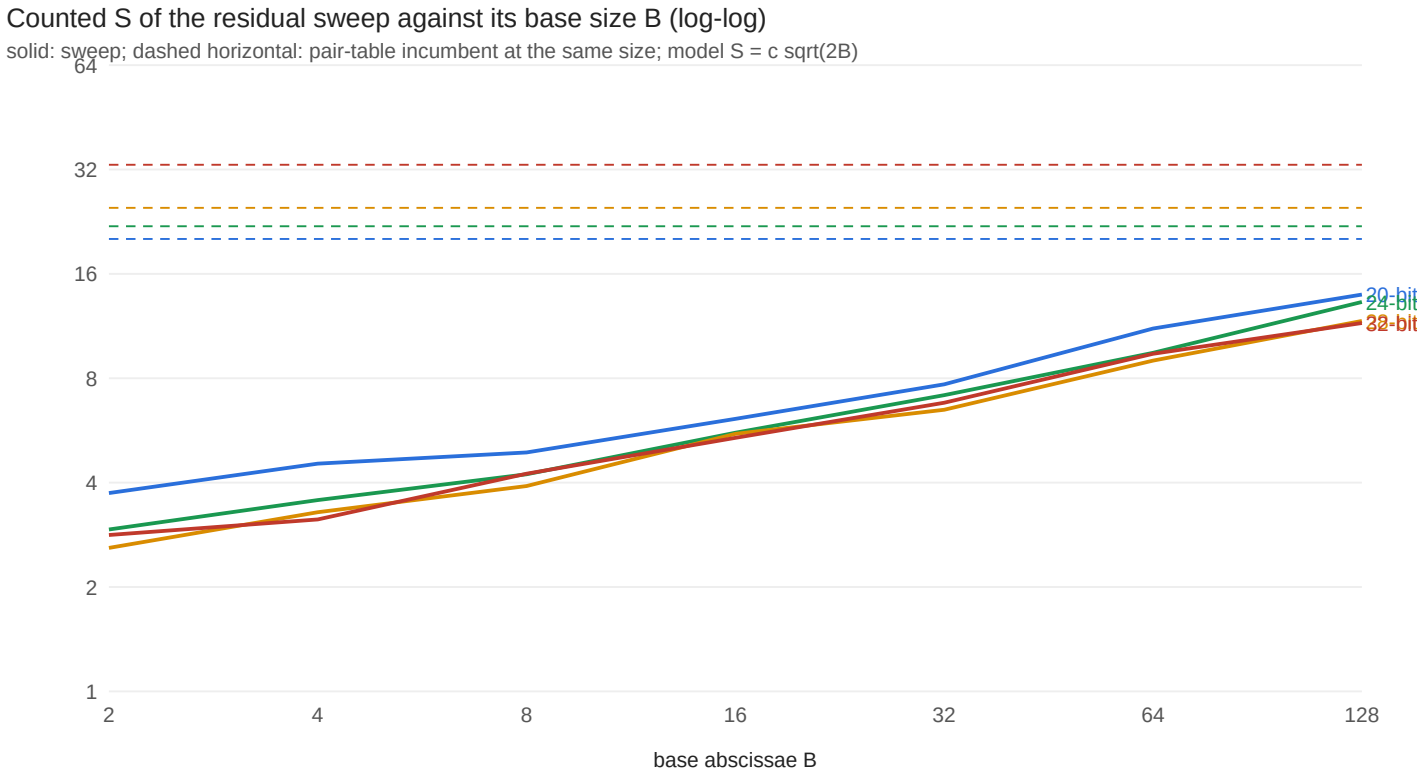
Single-large-prime residual sweep (2026-10-28)

Lead `ICP-residual-partial-relations`. Every residual R - P_i of a target is stored by abscissa; a collision of two residuals is a base-only relation. Three sessions per size in one binary; all arms alternated within each session.



Slope of log2 S against log2 B per size (model: +0.5): 20-bit +0.32, 24-bit +0.36, 28-bit +0.36, 32-bit +0.36.
Slope of log2 S against curve bits at fixed base (model: 0 for the sweep): sweep-b2 -0.035, sweep-b4 -0.043, sweep-b8 -0.018, sweep-b16 -0.014, sweep-b32 -0.017, sweep-b64 -0.020, sweep-b128 -0.025, pair-table (base ~ r^(1/3)) +0.058.

bits	arm	verified	mean S	S / rho.negation	paired S vs pair-table	residuals stored	table MB (est.)
20	rho-neg	36/36	1.14	1.0	nan	nan	nan
20	rho-lanes16	36/36	1.40	1.2	nan	nan	nan
20	pair-table-b128	36/36	20.17	17.7	1.000	nan	nan
20	sweep-b2	36/36	3.73	3.3	0.180	2163	0.12
20	sweep-b4	36/36	4.54	4.0	0.220	3111	0.17
20	sweep-b8	36/36	4.89	4.3	0.238	3627	0.19
20	sweep-b16	36/36	6.11	5.4	0.301	4904	0.26
20	sweep-b32	36/36	7.69	6.8	0.379	6512	0.35
20	sweep-b64	36/36	11.13	9.8	0.550	9931	0.53
20	sweep-b128	36/36	13.95	12.3	0.688	12736	0.68
24	rho-neg	36/36	1.03	1.0	nan	nan	nan
24	rho-lanes16	36/36	1.47	1.4	nan	nan	nan
24	pair-table-b256	36/36	21.96	21.4	1.000	nan	nan
24	sweep-b2	36/36	2.93	2.9	0.130	8069	0.43
24	sweep-b4	36/36	3.56	3.5	0.153	11120	0.59
24	sweep-b8	36/36	4.23	4.1	0.186	14191	0.76
24	sweep-b16	36/36	5.57	5.4	0.250	19613	1.05
24	sweep-b32	36/36	7.15	7.0	0.317	25877	1.38
24	sweep-b64	36/36	9.48	9.2	0.428	34955	1.87
24	sweep-b128	36/36	13.28	12.9	0.598	49643	2.65

28	rho-neg	36/36	1.03	1.0	nan	nan	nan
28	rho-lanes16	36/36	1.16	1.1	nan	nan	nan
28	pair-table-b512	36/36	24.80	24.1	1.000	nan	nan
28	sweep-b2	36/36	2.59	2.5	0.098	32337	1.73
28	sweep-b4	36/36	3.29	3.2	0.129	45871	2.45
28	sweep-b8	36/36	3.91	3.8	0.155	58091	3.10
28	sweep-b16	36/36	5.53	5.4	0.223	85152	4.55
28	sweep-b32	36/36	6.49	6.3	0.257	101680	5.43
28	sweep-b64	36/36	8.99	8.7	0.359	142517	7.61
28	sweep-b128	36/36	11.70	11.4	0.469	186667	9.97
32	rho-neg	36/36	1.02	1.0	nan	nan	nan
32	rho-lanes16	36/36	1.06	1.0	nan	nan	nan
32	pair-table-b1024	36/36	33.01	32.4	1.000	nan	nan
32	sweep-b2	36/36	2.82	2.8	0.080	131737	7.04
32	sweep-b4	36/36	3.13	3.1	0.092	162503	8.68
32	sweep-b8	36/36	4.25	4.2	0.127	233776	12.48
32	sweep-b16	36/36	5.38	5.3	0.162	305883	16.34
32	sweep-b32	36/36	6.80	6.7	0.203	392427	20.96
32	sweep-b64	36/36	9.41	9.2	0.282	548427	29.29
32	sweep-b128	36/36	11.55	11.3	0.343	675776	36.09

Findings

- **The sweep beats the pair-table recipe in the counted unit, by a lot.** At 32 bits S is 2.82 at B = 2, 5.38 at B = 16 and 11.55 at B = 128, against 33.01 for the incumbent (B = 1024) and 1.02 for rho.negation: the distance to rho falls from 32.4x to 2.8x. Every one of the 1008 sweep runs verified.
- **Its cost is a square root, and it is cheapest where there is least index calculus.** At fixed B the S does not grow with the curve (slope of log2 S against bits: -0.035, -0.043, -0.018, -0.014, -0.017, -0.020, -0.025), i.e. the work is c*sqrt(r) additions; S grows with the base as B^0.35 (model 0.5), so the smallest base wins. At B = 2 the base is four points and the search is a birthday collision of aG + bQ values.
- **The memory is the price the unit does not see.** The residual table holds 131737 entries at 32 bits for B = 2 (7.0 MB) and 675776 for B = 128 (36 MB); it grows like sqrt(r B). rho.negation keeps O(1) memory. S counts group additions only.
- **What this is.** A real, verified reduction of the cost of an index-calculus-pipeline arm and of its distance to rho; also the cost profile of a baby-step/giant-step collision search with a small signed base. The owner's rule that relabelling is not an improvement applies: this PR therefore does not put the sweep on tables A-C of the leaderboard and leaves that decision to the owner.
- **Unpriced native work.** Each residual costs a hash insert or probe that S does not charge. Priced at 1 GAE per probe the band keeps every ordering (sweep below the pair table, rho below the sweep).